

Learning: Naive Bayes

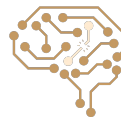
Arash Marioriyad

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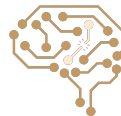
CE 417-Artificial Intelligence, Sharif Univ. of Tech.

Some of Slides have been adopted from Berkeley CS188 & SUT CE417



Up To Now ...

- Search
 - Uninformed & Informed & Local Search & CSP & Adversarial
- Uncertainty & Bayes Net
 - D-Separation & Inference
- Hidden Markov Model
 - Representation & Inference



Common Pattern Up To Now

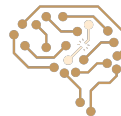
- No Data Involved
- A model of the world is already provided
- Question: how use this model to make optimal decisions?
- Assumption: some aspects of the problems are already known
 - In Search: Heuristics, Successor Function, Costs/Rewards, ...
 - In Bayes Nets: Conditional Probabilities, Network Structure, ...
 - In HMM: Transition Probabilities, Emission Probabilities, ...
- These aspects are called the **parameters** of the problem



Today

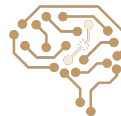
Learning

Infer Parameters from Data

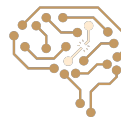
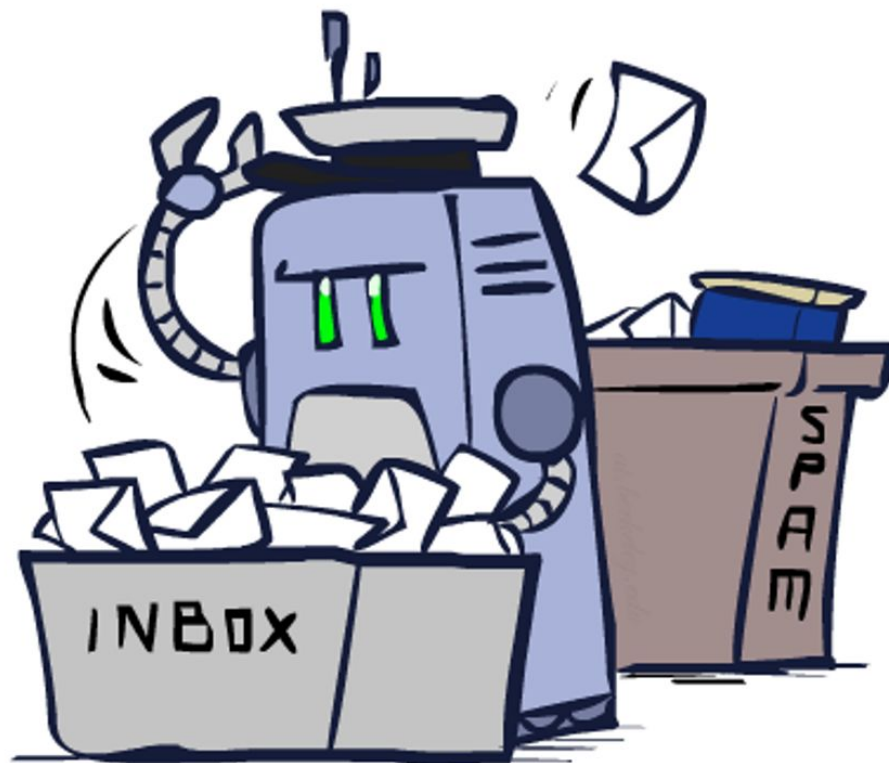


Machine Learning

- We do not have the model (parameters) in advance
- Question: how to acquire a model from data/experience?
 - Learning parameters (e.g. conditional probabilities)
 - Learning structure (e.g. BN graphs)
 - Learning hidden concepts (e.g. clustering)
- Today: model-based classification with Naive Bayes



Classification



Example: Spam Filter

- **Input:** an email
- **Output:** “spam” or “ham”
- **Data:** large collection of example emails, each labeled “spam” or “ham”
- **Note:** someone has to hand label all this data!
- **Goal:** to learn a model using the data, to predict labels of new, future emails
- **Features:** The attributes (words, text patterns, sender information, etc.,) that contribute in making an email ham or spam



Dear Sir.

First, I must solicit your confidence in this transaction, this is by virtue of its nature as being utterly confidential and top secret. ...

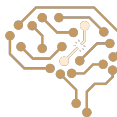


TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

99 MILLION EMAIL ADDRESSES FOR ONLY \$99



Ok, I know this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.



Example: Digit Recognition

- **Input:** images (pixels)
- **Output:** a digit 0-9
- **Data:** a large collection of example images, each labeled with a digit
- **Note:** someone has to hand label all this data!
- **Goal:** to learn a model using data, to predict labels of new, future digit images
- **Features:** The attributes used to make the digit decision
 - Pixels: example (6,8)=ON
 - Shape Patterns



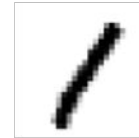
0



1



2



1



??

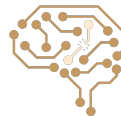


Example: Cat/Dog Classification

- **Input:** image of a cat or dog
- **Output:** a binary label (“cat” or “dog”)
- **Data:** a large collection of images, each labeled with “cat” or “dog”
- **Goal:** to learn a model using data, to predict labels of new, future images
- **Features:** Pixels



?

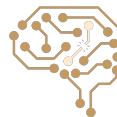


Other Classification Tasks

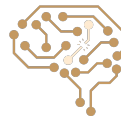
- **Classification:** given inputs x , predict labels (classes) y
- Examples:
 - Spam detection (input: document, classes: spam/ham)
 - OCR (input: images, classes: characters)
 - Medical diagnosis (input: symptoms, classes: diseases)
 - Automatic essay grading (input: document, classes: grades)
 - Fraud detection (input: account activity, classes: fraud / no fraud)



Classification is an important commercial technology!

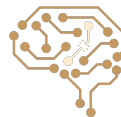


Model-Based Classification



Model-Based Classification

- Model-based approach: Build a model (e.g. **Bayes' net**) where both the label and features are random variables
- **Features (attributes)** are evidence (observed) variables
- Query for the distribution of the label (query variable) conditioned on the features (observed variables)
- Challenges:
 - **What structure should the BN have?**
 - **How should we learn its parameters (CPTs)?**

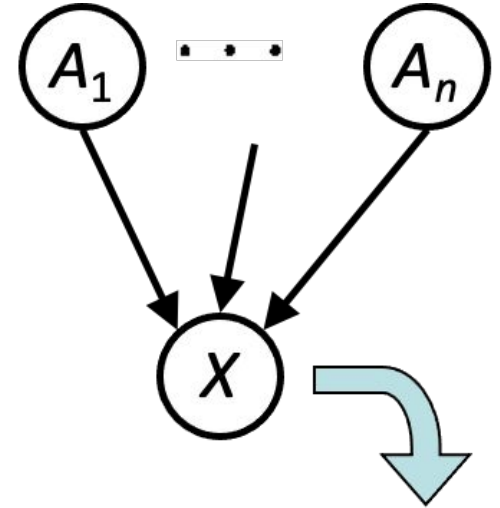


Recall: Bayes Net

- A directed, acyclic graph (DAG)
- A set of nodes, one per variable
- A conditional distribution for each node

$$P(X|a_1 \dots a_n)$$

- CPT: conditional probability table

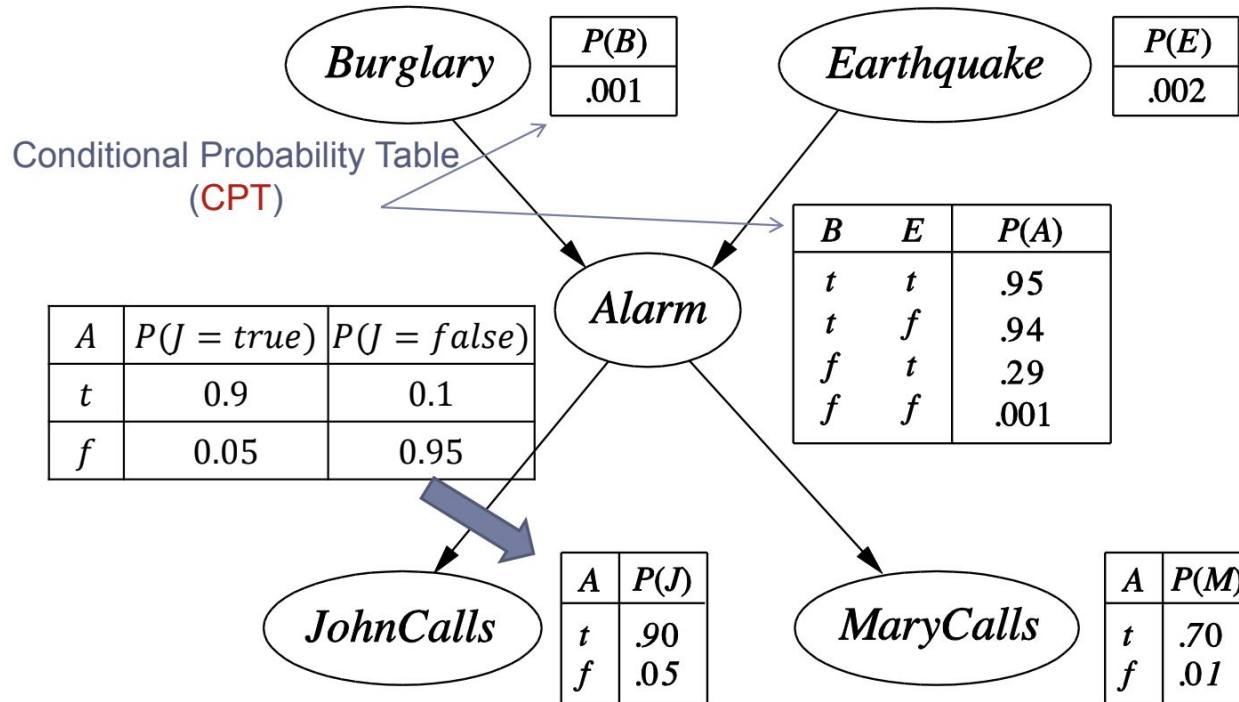


$$P(X|A_1 \dots A_n)$$

A Bayes net = Topology (graph) + Local Conditional Probabilities



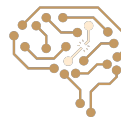
Bayes Net: Burglary Example



Probabilities in BNs

- Bayes' nets implicitly encode joint distributions.

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



Bayes Net For Classification: Nodes

- **Class node (Y):**
 - Represents the target label or category we want to predict.
- **Feature (attribute) nodes ($X_1, X_2, \dots, X_n$):**
 - Observed variables describing the instance
- Sometimes feature nodes are shown as $F_1, F_2, \dots, F_n$



Bayes Net For Classification: Edges

- **Edges between class node and features:**
 - $Y \rightarrow X_i$: class causes the observed attributes.
- **Dependency edges among features:**
 - Edges between X_i and X_j to encode relations between features



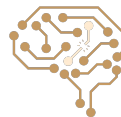
Bayes Net For Classification: Goal

- Use a probabilistic graphical model to compute:

$$\arg \max_y P(Y = y \mid x_1, \dots, x_n) = \arg \max_y \frac{P(Y = y, x_1, \dots, x_n)}{\sum_{y'} P(Y = y', x_1, \dots, x_n)}.$$

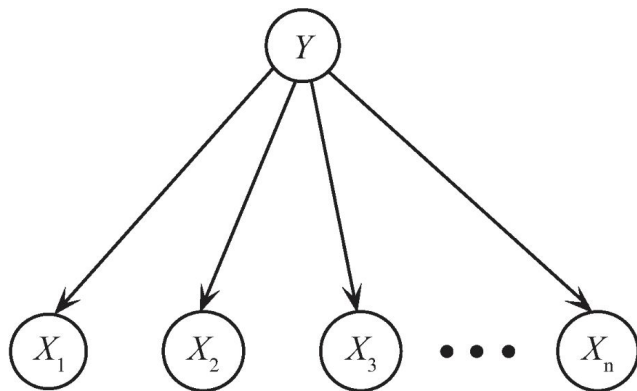
- And the joint distribution is:

$$P(Y, X_1, \dots, X_n) = P(Y) \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i))$$

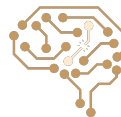


Naive Bayes

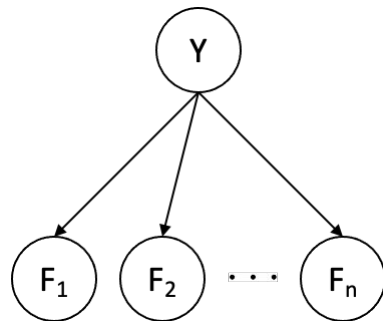
- Naive Assumption: all features are **independent** of each other
- Edges are only from the class variable to the feature variables



$$P(Y, X_1, \dots, X_n) = P(Y) \prod_{i=1}^n P(X_i | Y)$$



Naïve Bayes for Digits (Grayscale Images)



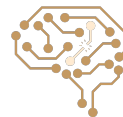
- Simple digit recognition version:
 - One feature (variable) F_{ij} for each grid position $\langle i,j \rangle$
 - Feature values are “on” or “off”, based on whether intensity is more or less than 0.5 in underlying image
 - Each input maps to a feature vector, e.g.
 - Here: lots of features, each is binary valued

1

$\rightarrow \langle F_{0,0} = 0 \ F_{0,1} = 0 \ F_{0,2} = 1 \ F_{0,3} = 1 \ F_{0,4} = 0 \ \dots F_{15,15} = 0 \rangle$

- Naïve Bayes model:

$$P(Y|F_{0,0} \dots F_{15,15}) \propto P(Y) \prod_{i,j} P(F_{i,j}|Y)$$



So ...

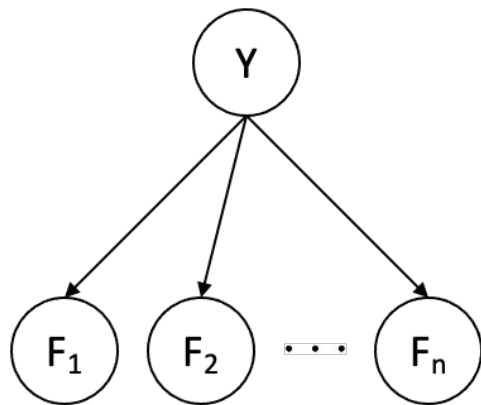
- A Naive Bayes model:

$|Y|$ parameters

$$P(Y, F_1 \dots F_n) = P(Y) \prod_i P(F_i | Y)$$

$|Y| \times |F|^n$ values

$n \times |F| \times |Y|$
parameters

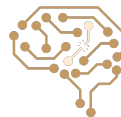


- We only have to specify how each feature depends on the class
- Total number of parameters is linear in n
- Model is very simplistic, but often works anyway



What Must We Learn in Naïve Bayes?

- Estimates of local conditional probability tables:
 - $P(Y)$, the prior over labels
 - $P(F_i|Y)$ for each feature (evidence variable)
- These probabilities are collectively called the parameters of the model and denoted by θ
- Up until now, we assumed these appeared by magic, but...
- But they typically come from training data



Inference for Naïve Bayes

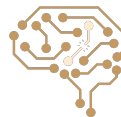
- **Goal:** compute posterior distribution over label variable Y
- **Step 1:** get joint probability of label and evidence for each label

$$P(Y, f_1 \dots f_n) = \begin{bmatrix} P(y_1, f_1 \dots f_n) \\ P(y_2, f_1 \dots f_n) \\ \vdots \\ P(y_k, f_1 \dots f_n) \end{bmatrix} \Rightarrow \begin{bmatrix} P(y_1) \prod_i P(f_i|y_1) \\ P(y_2) \prod_i P(f_i|y_2) \\ \vdots \\ P(y_k) \prod_i P(f_i|y_k) \end{bmatrix}$$

$\xrightarrow{+}$

- **Step 2:** sum to get probability of evidence
- **Step 3:** normalize by dividing Step 1 by Step 2

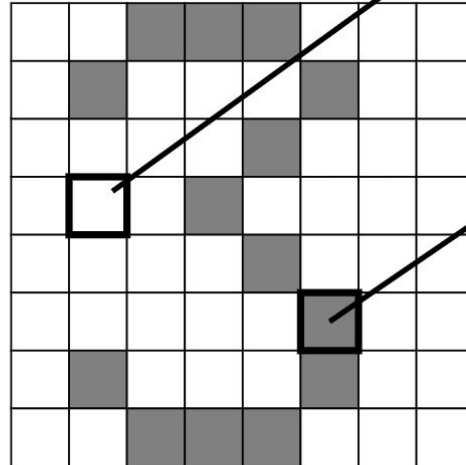
$$\frac{\begin{bmatrix} P(y_1) \prod_i P(f_i|y_1) \\ P(y_2) \prod_i P(f_i|y_2) \\ \vdots \\ P(y_k) \prod_i P(f_i|y_k) \end{bmatrix}}{P(f_1 \dots f_n)} \Rightarrow P(Y|f_1 \dots f_n)$$



Example: Conditional Probabilities

$P(Y)$

1	0.1
2	0.1
3	0.1
4	0.1
5	0.1
6	0.1
7	0.1
8	0.1
9	0.1
0	0.1

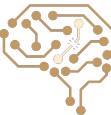


$P(F_{3,1} = on|Y)$

1	0.01
2	0.05
3	0.05
4	0.30
5	0.80
6	0.90
7	0.05
8	0.60
9	0.50
0	0.80

$P(F_{5,5} = on|Y)$

1	0.05
2	0.01
3	0.90
4	0.80
5	0.90
6	0.90
7	0.25
8	0.85
9	0.60
0	0.80



Naïve Bayes for Text

- Features: W_i is the word at position i
- As before: predict label (spam vs. ham) conditioned on feature variables (words)
- As before: assume features are conditionally independent given label (not necessarily a good assumption)

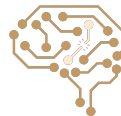
$$P(Y, W_1 \dots W_n) = P(Y) \prod_i P(W_i | Y)$$

*Word at position
 i , not i^{th} word in
the dictionary!*



Bag of Words (BoW)

- Usually, each variable gets its own conditional probability distribution $P(F|Y)$
- In a bag-of-words model:
 - Each position is identically distributed
 - All positions share the same conditional probs $P(W|Y)$
 - Model is insensitive to word order or reordering
 - Why make this assumption?



Example: Spam Filtering

Model: $P(Y, W_1 \dots W_n) = P(Y) \prod_i P(W_i|Y)$

What are the parameters?

$P(Y)$

ham	: 0.66
spam	: 0.33

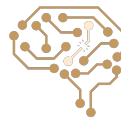
$P(W|\text{spam})$

the	:	0.0156
to	:	0.0153
and	:	0.0115
of	:	0.0095
you	:	0.0093
a	:	0.0086
with:		0.0080
from:		0.0075
...		

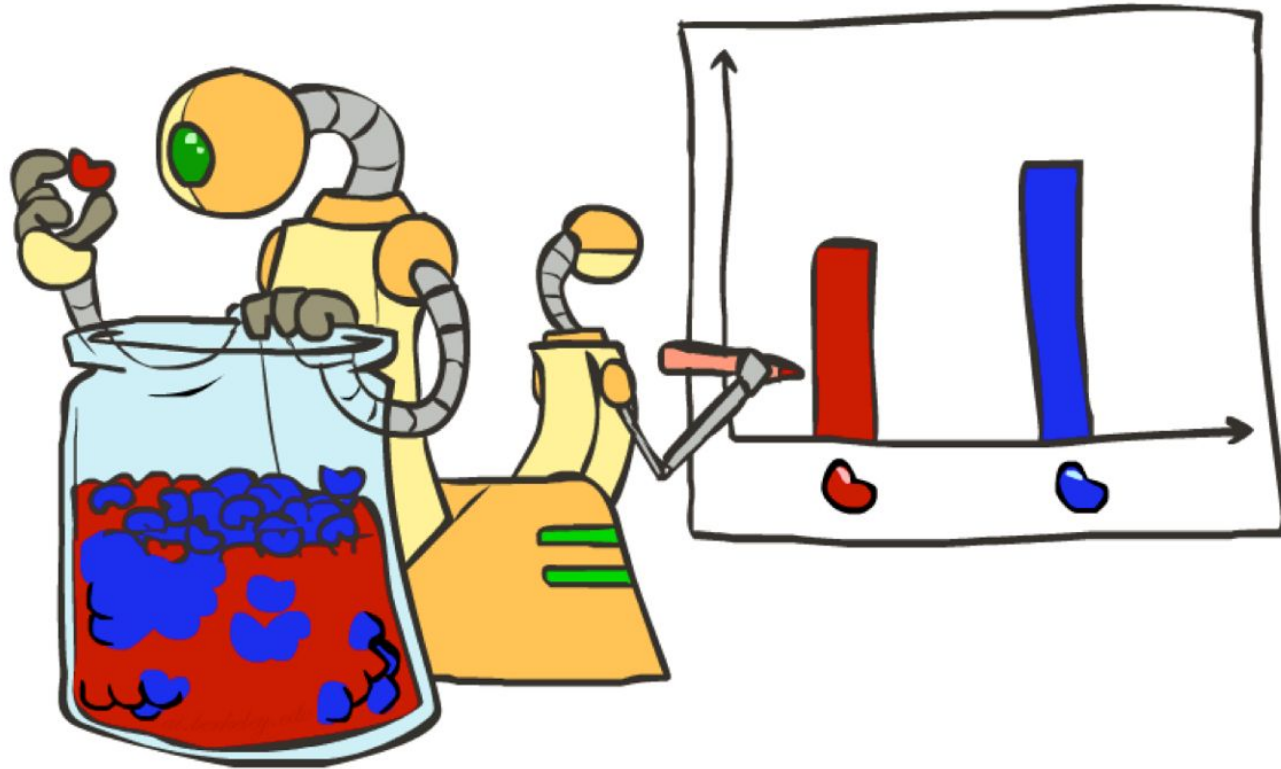
$P(W|\text{ham})$

the	:	0.0210
to	:	0.0133
of	:	0.0119
2002:		0.0110
with:		0.0108
from:		0.0107
and	:	0.0105
a	:	0.0100
...		

Where do these tables come from?



Parameter Estimation



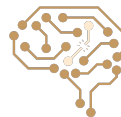
How to Learn Parameters From Data?

- Algorithm: **Maximum Likelihood Estimation (MLE)**
- Two types of parameters in Naive Bayes
- **Class Prior:**

$$P(Y = y) = \frac{\text{number of training examples with } Y = y}{N}$$

- **Conditional Probabilities:**

$$P(X_i = x \mid Y = y) = \frac{\#\{X_i = x, Y = y\}}{\#\{Y = y\}}$$



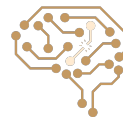
Maximum Likelihood Estimation

$$\theta_{ML} = \arg \max_{\theta} P(\mathbf{X}|\theta)$$

$$= \arg \max_{\theta} \prod_i P_{\theta}(X_i)$$

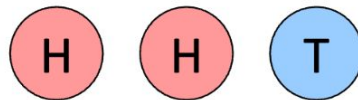


$$P_{ML}(x) = \frac{\text{count}(x)}{\text{total samples}}$$



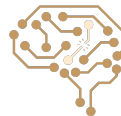
Maximum Likelihood Estimation: Example

- Estimating the distribution of a random variable
 - E.g., here is a coin; what is the probability θ of heads?
- Evidence $\mathbf{d} = d_1, \dots, d_N$
 - E.g., three independent coin tosses $d_1=\text{heads}$, $d_2=\text{heads}$, $d_3=\text{tails}$
- Likelihood: probability of the evidence $P(d_1, \dots, d_N; \theta)$
 - E.g., $P(d_1=\text{heads}, d_2=\text{heads}, d_3=\text{tails}; \theta) = \theta^2(1-\theta)$
- Maximum likelihood: What value θ_{ML} maximizes the likelihood?
- Log likelihood: $L(\mathbf{d}; \theta) = \log P(\mathbf{d}; \theta)$
 - E.g., $L(\mathbf{d}; \theta) = 2 \log \theta + \log(1-\theta)$
- θ_{ML} also maximizes the log likelihood (why?) and it's easier to differentiate (why?)
- $\partial L / \partial \theta = 2/\theta - 1/(1-\theta) = 0$
- $\theta_{ML} = 2/3$
- For h heads and t tails, $\theta_{ML} = h/(h+t)$

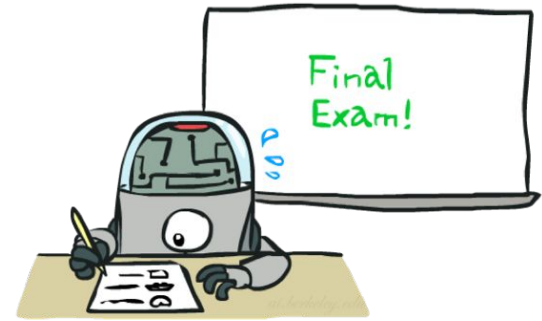
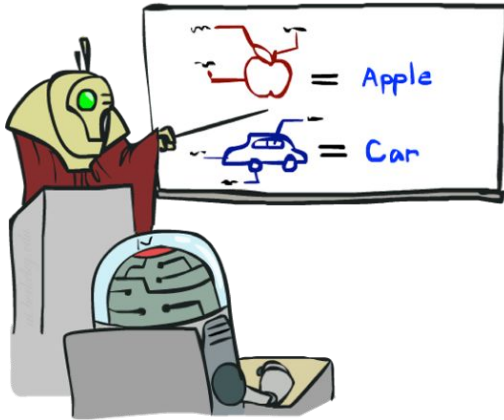


Beyond Naive Bayes: Causal Structure Learning

- Not a topic covered in this course!
- **Goal:** Learn the graph structure (which variables cause or influence others) directly from data.
- Two main approaches:
 - **Score-based methods:** Search over possible DAGs and choose the one with highest score (e.g., BIC, MDL).
 - **Constraint-based methods:** Use conditional independence tests (e.g., PC algorithm) to infer edges.
- **Output:** A directed acyclic graph (DAG) that reflects probable causal relationships among variables.

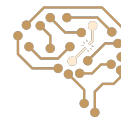
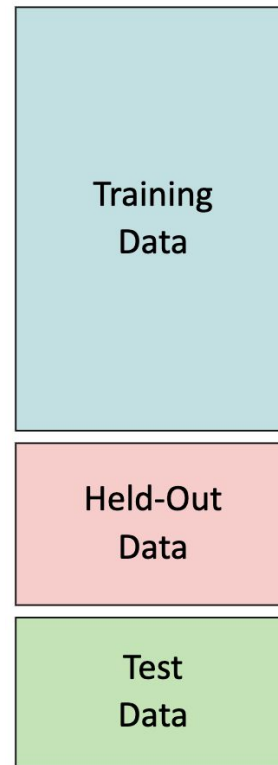


Machine Learning Concepts



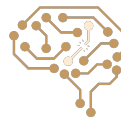
Data

- Three Types of Datasets (Labeled Instances):
 - Training Set
 - Validation (Held-out) Set
 - Test Set



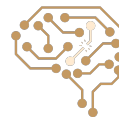
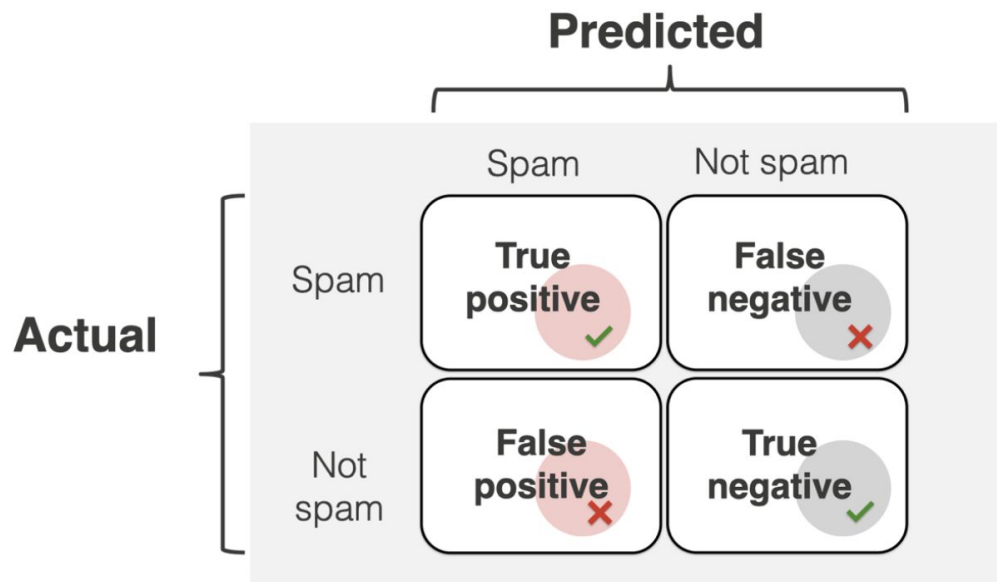
Process

- We want the model that does best on the true distribution
- But we do not know the true distribution (we just have datasets)
- So learn model parameters (e.g. probabilities) on training set
- Tune hyperparameters on held-out set
- Compute performance metrics (e.g. accuracy) of test set
- Very important: never “peek” at the test set!



Performance Metrics

- **Evaluation** = Compute performance metrics over a dataset
- For binary classification, we have 2X2 **confusion matrix**:



Evaluation (cont.)

- Accuracy

- Number of correct predictions from the entire data set:
 $(TP + TN) / (TP + FP + TN + FN)$

- Precision

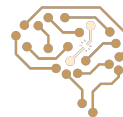
- Number of correct positive predictions from total positive predictions:
 $TP / (TP + FP)$

- Recall

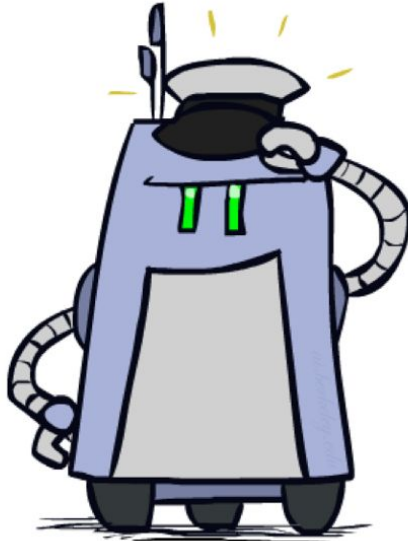
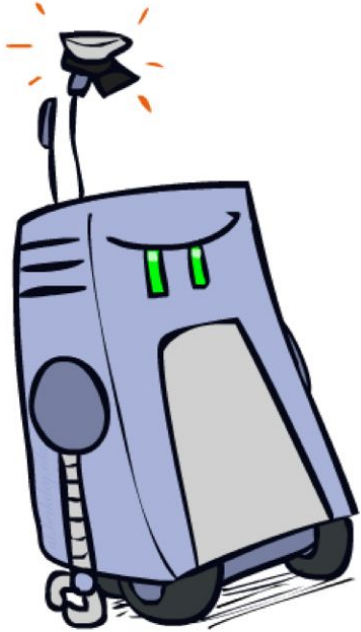
- Number of correct positive predictions from the actual positive samples:
 $TP / (TP + FN)$

- F-score

- Harmonic mean of Precision and Recall: $2TP / (2TP + FP + FN)$



Generalization and Overfitting



Overfitting Example: Digit Recognition

$P(\text{features}, C = 2)$

$P(C = 2) = 0.1$

$P(\text{on}|C = 2) = 0.8$

$P(\text{on}|C = 2) = 0.1$

$P(\text{off}|C = 2) = 0.1$

$P(\text{on}|C = 2) = 0.01$

$P(\text{features}, C = 3)$

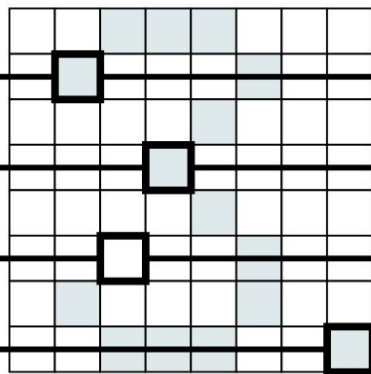
$P(C = 3) = 0.1$

$P(\text{on}|C = 3) = 0.8$

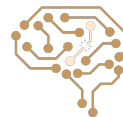
$P(\text{on}|C = 3) = 0.9$

$P(\text{off}|C = 3) = 0.7$

$P(\text{on}|C = 3) = 0.0$



2 wins!!



Overfitting Example: Text Classification

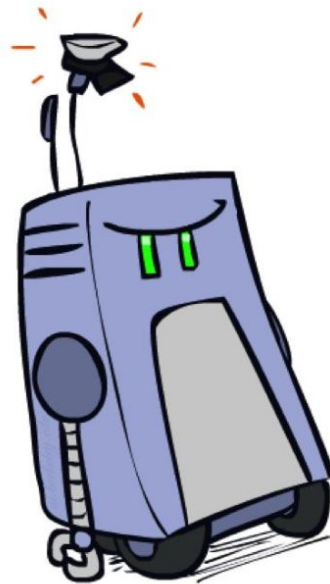
$$\frac{P(W|\text{ham})}{P(W|\text{spam})}$$

south-west	:	inf
nation	:	inf
morally	:	inf
nicely	:	inf
extent	:	inf
seriously	:	inf
...		

$$\frac{P(W|\text{spam})}{P(W|\text{ham})}$$

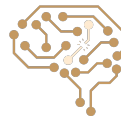
screens	:	inf
minute	:	inf
guaranteed	:	inf
\$205.00	:	inf
delivery	:	inf
signature	:	inf
...		

What went wrong here?



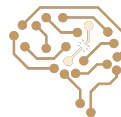
Overfitting

- Relative frequency parameters will overfit the training data!
 - Just because we never saw a 3 with pixel (15,15) on during training doesn't mean we won't see it at test time
 - Unlikely that every occurrence of "minute" is 100% spam
 - Unlikely that every occurrence of "seriously" is 100% ham
 - What about all the words that don't occur in the training set at all?
 - In general, we can't go around giving unseen events zero probability

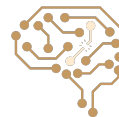
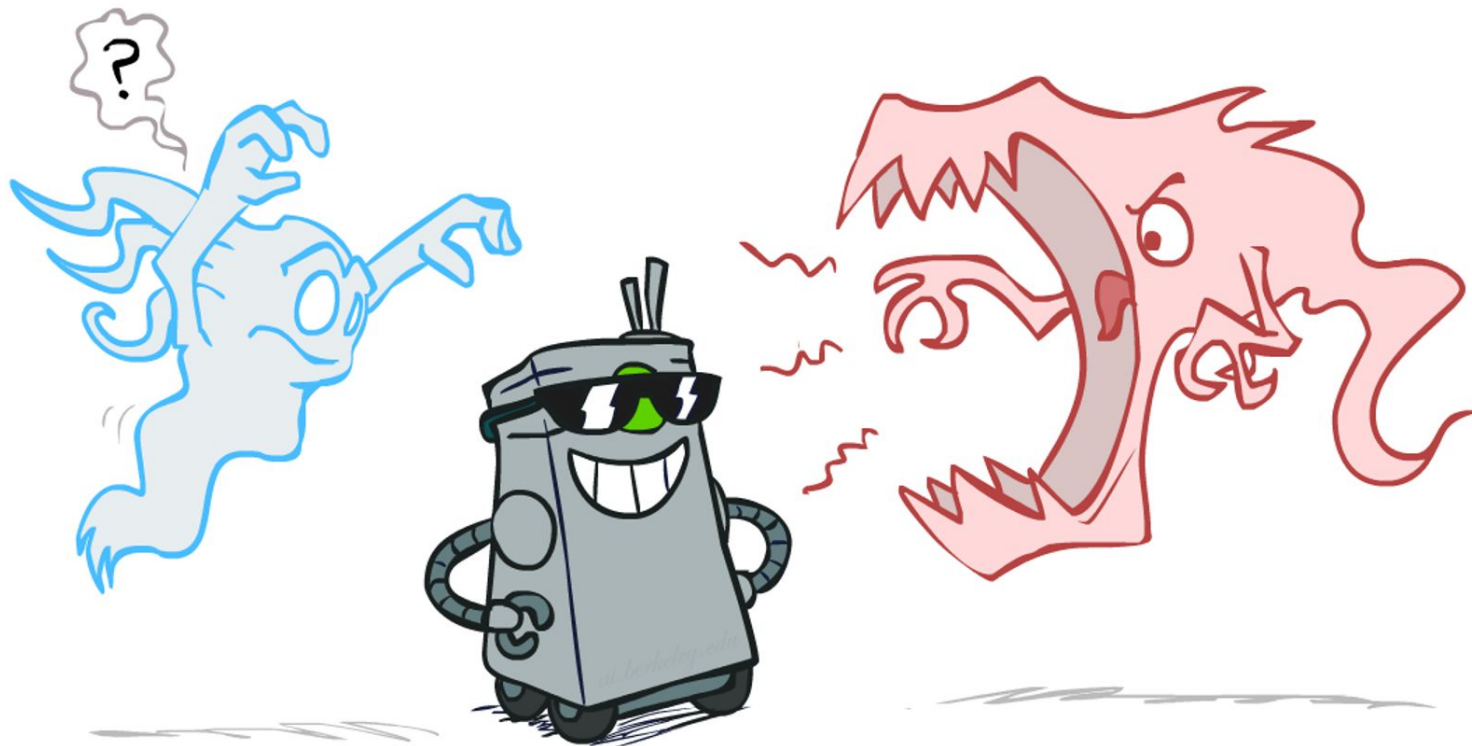


Generalization

- Perform well on unseen data, not just the training examples
- Failure mode = Overfitting
 - the model fits training frequencies too closely and assigns unreliable probabilities to new cases
- Solution in Naive Bayes: Smoothing

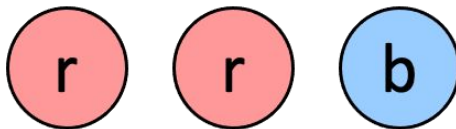


Smoothing



Laplace Smoothing

- Pretend you saw every outcome once more than you actually did

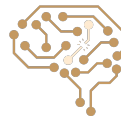


$$P_{LAP}(x) = \frac{c(x) + 1}{\sum_x [c(x) + 1]}$$

$$= \frac{c(x) + 1}{N + |X|}$$

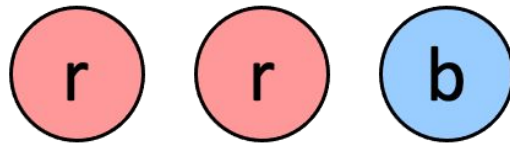
$$P_{ML}(X) = \left\langle \frac{2}{3}, \frac{1}{3} \right\rangle$$

$$P_{LAP}(X) = \left\langle \frac{3}{5}, \frac{2}{5} \right\rangle$$



Laplace Smoothing in General

- Pretend you saw every outcome k extra times



$$P_{LAP,k}(x) = \frac{c(x) + k}{N + k|X|}$$

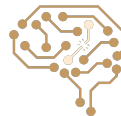
- What's Laplace with $k = 0$?
- k is the strength of the prior
- Laplace for conditionals:

$$P_{LAP,0}(X) = \left\langle \frac{2}{3}, \frac{1}{3} \right\rangle$$

$$P_{LAP,1}(X) = \left\langle \frac{3}{5}, \frac{2}{5} \right\rangle$$

$$P_{LAP,100}(X) = \left\langle \frac{102}{203}, \frac{101}{203} \right\rangle$$

$$P_{LAP,k}(x|y) = \frac{c(x, y) + k}{c(y) + k|X|}$$

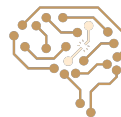


Linear Interpolation

- In practice, Laplace often performs poorly for $P(X|Y)$
- Also get the empirical $P(X)$ from the data
- Make sure the estimate of $P(X|Y)$ is not too different from the empirical $P(X)$

$$P_{LIN}(x|y) = \alpha \hat{P}(x|y) + (1.0 - \alpha) \hat{P}(x)$$

- What if α is 0? Or 1?



Real NB: Smoothing

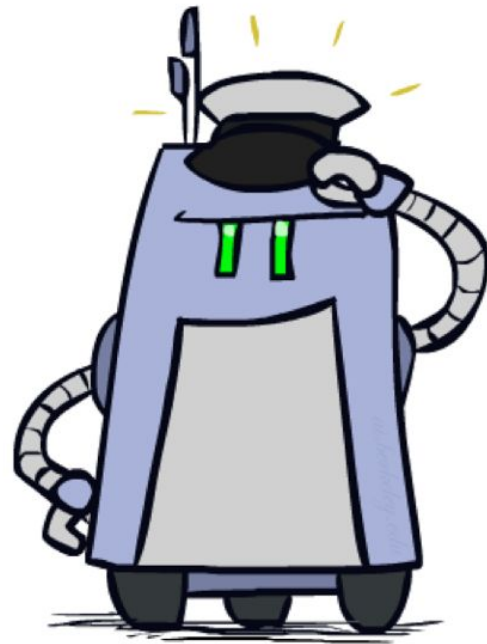
$$\frac{P(W|\text{ham})}{P(W|\text{spam})}$$

helvetica	:	11.4
seems	:	10.8
group	:	10.2
ago	:	8.4
areas	:	8.3
...		

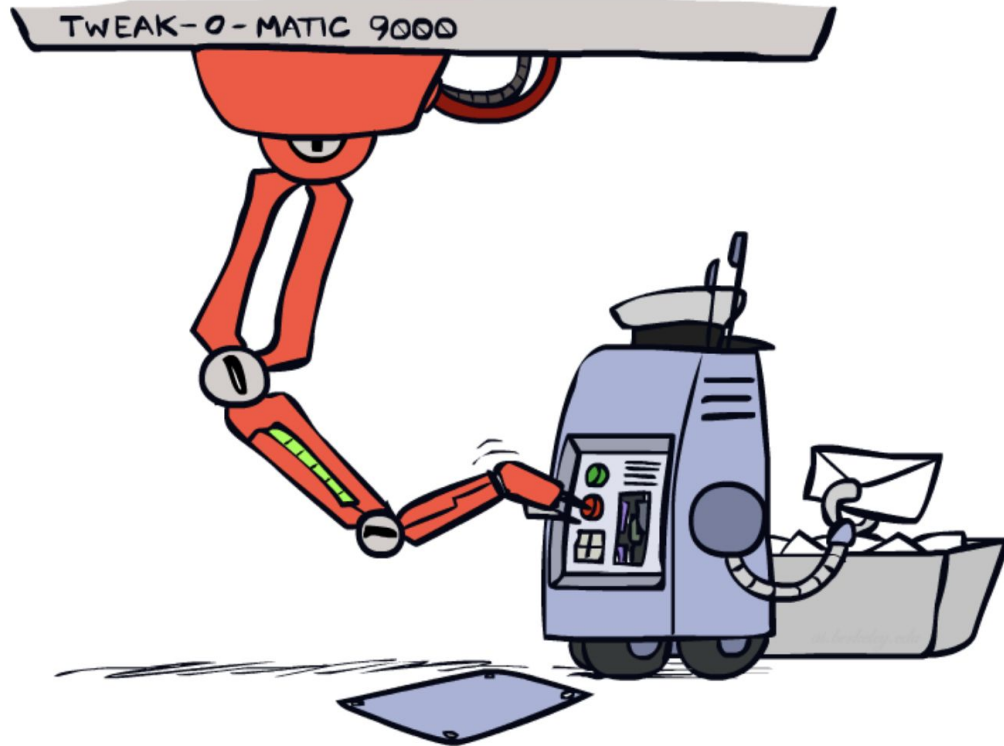
$$\frac{P(W|\text{spam})}{P(W|\text{ham})}$$

verdana	:	28.8
Credit	:	28.4
ORDER	:	27.2
	:	26.9
money	:	26.5
...		

Do these make more sense?



Hyperparameter Tuning



Tuning on Held-Out (Validation) Data

- Now we've got two kinds of unknowns
 - Parameters: the probabilities $P(X|Y)$, $P(Y)$
 - Hyperparameters: k in Laplace smoothing and α in linear Interpolation
- What should we learn where?
 - Learn parameters from training data
 - Tune hyperparameters on validation data. Why?
 - For each value of the hyperparameters, train and test on the held-out (validation) data
 - Choose the best hyperparameter value and do a final test on the test data



Next Session: Decision Trees

